

CONCEIVING FOREST MANAGEMENT AS PROVIDING FOR CURRENT AND FUTURE SOCIAL VALUE

JAMES J. KENNEDY

Forest Resources Department, Utah State University, Logan, UT 84322 (U.S.A.)

Visiting Fullbright Scholar at Trinity College, Dublin, Ireland

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ABSTRACT

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This paper emphasizes the role of planning and management in providing for forest social values to current and future citizens. Taking a western-world orientation, it examines how forest values originate in society and are expressed to forest managers.

For most of the last 200 years, the social needs of public and private forests in the western world were predominately utilitarian and their immediate social value was well expressed in market prices. As these countries urbanized, romantic and symbolic forest values increased. Today the social values of forest recreation, landscapes and non-game wildlife are of increased importance. These forest social values are usually expressed by social and political systems.

Since utilitarian social values are often in conflict with romantic/symbolic forest values today, foresters increasingly are placed in the role of conflict managers. In what they do, and do not do, foresters can dampen or exacerbate forestry social conflict.

INTRODUCTION

At the beginning of America's environmental movement Toffler (1970) published a book called *Future Shock*, where the world was presented as changing at a rapid rate. He probably would not be surprised that I and many American foresters have been shocked by the rapid changes in our professional images and roles since 1970 (Reidel, 1971; Behan, 1975). One reason for this shock was foresters' traditional view of themselves managing and protecting *resource things* (i.e., objects like trees or game animals), rather than managing these resources as objects of changing forest social values (including non-consumptive and symbolic values).

Another reason why many American foresters were shocked by societal changes in the last 20 years stems from their professional pride and cohesiveness, born of successes in the conservation era (1900–1960). At that

time, foresters had more clarity of purpose and wider respect (and/or benign disinterest) of the public than now. American society has become more urban and diverse in the last three generations. Fifty years ago society valued forests primarily for their soil, water and wood benefits that insured their dreams of increased material progress and national security could endure from generation to generation (Hays, 1959). The conservation movement also occurred in an era when science and professional expertise were widely revered (Frankel, 1969) and incorporated into public trust of government and private forest land management.

Environmental movements that stressed non-utilitarian and romantic/idealistic social values imposed new agendas on traditional forestry and on foresters in America and much of the urbanizing western world. Forest management has become even more challenging because of the conflicts between groups supporting consumptive and non-consumptive/symbolic forest values. Forest managers in North America, Sweden, the United Kingdom and New Zealand have had to cope with such future shock (Beale, 1970; Hopkins, 1970; Hagenstein, 1971; Kennedy, 1981).

Accepting the definition of forestry as the management of forest resources to provide a satisfactory amount and mix of social values for clients living, while protecting these values and use options for future generations, this paper will (1) explore how forest social values originate and are expressed to foresters; (2) propose some factors underlying national conflict over these values; and (3) comment on the evolving image and roles of foresters in a world of diverse, changing and oft-conflicting forest values. I will focus on foresters as one important type of forest manager, recognizing that forest management today often includes landscape architects, wildlife biologists, hydrologists and other professionals.

ORIGIN AND EXPRESSION OF FOREST SOCIAL VALUES

Rokeach's (1973) concept of value as "an enduring belief that a specific mode of conduct or end-state of existence is personally or socially preferable to an opposite or converse mode . . ." recognizes human values as relatively stable beliefs. People employ such beliefs to judge forest practices (e.g., herbicide spraying, clearcutting or prescribed burning) and associated end-states (e.g., increased wood productivity, more local employment, or less forest plant and wildlife diversity). These beliefs (values) have both personal and social desirability aspects (Brown, 1984), plus short- and long-run time dimensions. For example, a person believes that fast-growing, intensively managed pine plantations are aesthetically pleasing and provide important employment options and long-run community socio-economic stability. Such human values, although stable, do evolve over time (e.g., the evolution of 1910 Protestant work-ethic attitudes towards enjoyment of leisure to current American beliefs, expressed in increased forest recreation).

It is proposed that such forest social value exist in interrelated sets of many systems. Four major systems are conceptualized in Fig. 1 as: (1) the *environmental/natural resource system* of natural biosphere elements, such as forest and other natural (and environmental) resources; (2) the *social/cultural system* of human (individual and group) attitudes, values, behavior, institutions and technology; (3) the *economic system* that includes certain types of human behavior, technology and institutions relative to the utilization of land, labor and capital resources; and (4) the *political system* of social attitudes/values and behavior that are explicitly defined and regulated by policy, legislation and legal/police force. As Fig. 1 illustrates, forest resources (in the environmental system) affect and are affected by all other systems. That is, the four systems all intersect (interact) with one another.

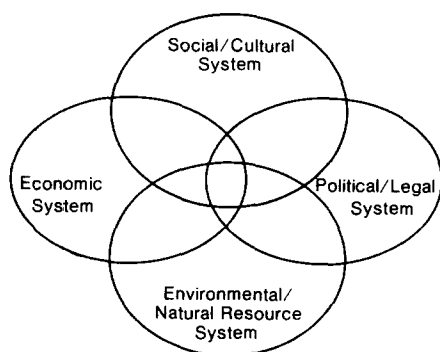


Fig. 1. Systems originating and expressing forest values.

Forest values *originate* in only one system, the social/cultural. Forest values are *expressed* (or communicated) by three systems: the economic, social, and political. In this western-world conceptualization, the physical/environmental system itself neither originates nor expresses forest values. It is only through human interaction with physical resources that forest values originate and are expressed. We will explore these concepts in more detail.

The origin of forest social values

The Judeo-Christian and Greek (and possibly male-oriented; Grey, 1979) heritage of western culture views nature as having no inherent value. Nature or forests are valued only if they fulfill human needs (Passmore, 1974; Foley, 1977). Views, characteristic of Buddhists or American Indians, that accept inherent nature values independently of their utilitarian worth are gaining credibility in western society. But Buddhist and all other forest value orientations are devices of the human mind, shaped by interactions of culture and environment, and *originating* in the social/cultural system (Fig. 1).

Maslow (1954) offers a model of need-seeking fulfillment of any human endeavor — including the consumptive or non-consumptive use of forest resources. His hierarchy of human “felt and expressed needs” (Mercer, 1973) ranges from lower-level subsistence to higher-level esteem and self-actualization needs. For example, big game hunting in North American forests may once have primarily satisfied subsistence (meat) needs. Today such hunting satisfies needs higher in Maslow’s hierarchy, such as social needs (Kennedy, 1974) or status and self-identity (Hendee, 1974). Maslow’s hierarchy defines the source of society’s basic need for game meat, trophy antlers, bird watching, hardwood furniture, fire wood, or camping in a wilderness area (Hendee et al., 1968; Tocher, 1971). From these human needs, in the social/cultural system, originate all forest (or other human) values.

Does this mean that a unique tree like the North American redwood (*Sequoia sempervirens*) has no value in itself? Yes, it does! Redwood values are not genetically “imprinted” in the human mind. They are learned socially. Valuing redwoods as a source of wood and employment, or as national heirloom trees, is socially learned attitude formation.

Of course forest values are not formed in isolation within the social/cultural system, but as that system interacts with environmental and other systems. Agricultural/rural societies tend to have different needs of forests than do urban societies, usually resulting in different forest perceptions, values, and uses. Most modern forest controversies are conflicts between consumptive values and nonconsumptive/symbolic values about human relationships with and use of nature (e.g., use and management of North American redwood or New Zealand native forests). Consumptive forest values are commonly associated with agricultural/rural society and non-consumptive values with urban society. But, as discussed in a later section, such values association is not clearly established.

The expression or communication of forest social values

Brown’s (1984) comprehensive paper on the use of values in natural resource management recognizes that the “held values” of individuals and groups affect how they “assign values” in choice behavior. Only when internal, “held” values are *expressed* in various systems can natural resource managers begin to understand what values individuals and groups “assign” forest practices and outputs. Common *social* system expressions of forest values are responses to questionnaires, number and type of people using recreational areas, letters to the editor, or a government forester (and her family) being shunned in a local community. The *economic* system expresses forest values in the consumption and price of wood products, expenditures in pursuit of recreation or sale of summer home sites. Legislation, budget allocation and law suits are value expressions of the *political* system.

Historically foresters have been conditioned to understand and respond to the social value communicated by the economic system, as the primary means of current generations expressing their forest values. There were still the values of future generations to consider — these often not well expressed by the economic system. Professional forestry norms (e.g., sustained yield) and political systems (e.g., laws and policies) attempted to provide for forest values of society unborn. Behan (1977) has observed that these professional forester norms often dominated application of basic economic efficiency logic in National Forest management, and resisted adaptive behavior to new forest social values (Twight, 1983).

The economic system can be conceptualized as an interaction of the environmental system with technology (supply) and preferences (demand) of the social/cultural system, resulting in forest social values expressed by market prices. The political/legal system is an extension of the social system that expresses forest values of current and future generations through legislation, subsidies or tax incentives.

Many important social values are expressed by the prices people are willing to pay for forest goods and services. The United States in the first half of this century, for example, was an agricultural and immigrant society attempting to industrialize and provide a good, sustainable standard of living for its citizens. The economic system effectively expressed the immediate and predominantly utilitarian forest values of that society. Of course, there was concern over the myopia of laissez-faire capitalistic forestry in the United States and Europe, exhibited in harvesting without adequate regeneration, or logging impacts on flooding, avalanche and forest fire risks. These failures of the economic system provided adequate justification for the socio-political expression of changing forest values in the American conservation movement. Concern for wildlife, loss of wildlands and the American frontier heritage, and other non-product values also contributed to the socio-political force of this conservation movement (Gilligan, 1954; Nash, 1967). Today, as in the past, the value Americans have for black walnut wood (*Juglans nigra*) or the Japanese for Port Orford cedar (*Chamaecyparis lawsoniana*), as expressed in the price of these forest products, is still a powerful force directing the behavior of American foresters managing public and private forestland.

As non-product benefits like recreation, landscape amenities or non-game wildlife species became more important to urban citizens of western countries in the last 20 years, non-economic systems became increasingly influential in expressing widening forest values. The Wilderness Act (1964) and the National Environmental Policy Act (1970) were political expressions of such values, active in American forest policy and management for decades. New Zealand's Native Forest Act (1976) or Sweden's Preservation of Beech Forests and Forestry Act (1979) are similar socio-political expressions of expanding forest values. The National Environmental Policy Act also required public involvement, so that Americans could more easily

express their forest/environmental values in social and political systems. This was to counter-balance the power of market prices (economic system) and the tradition of government agencies dominated by professional monocultures (e.g., civil engineers in the Army Corps of Engineers, or foresters in the USDA Forest Service). In case federal natural resource agencies did not respond to social values expressed in hearings and other public involvement, much recent legislation allows the government to be taken to court — a form of political/legal value communication that is difficult to ignore.

Initial cognitive and normative barriers may have existed with foresters understanding and responding to widening forest values expressed in social and political systems. Cognitively, foresters were not well prepared by education or experience to understand and respond to social, political and legal expressions of forest values. Most foresters educated in the European tradition take courses in economics and policy (the latter generally from a historical—descriptive perspective). Few courses educate foresters to understand and appreciate the legal, social and political dynamics of modern forest management.

Normative barriers may also have inhibited some foresters in listening and responding to forest values expressed by public hearings, special interest groups or legal action. Foresters had decades experience with values communicated through prices from the economic system. These indications of value were quantifiable. They could also be viewed as objective, unbiased expressions of a “free enterprise” system. Politics and public hearings express social values that are difficult to quantify and evaluate. They can be viewed as emotional, self-interested expressions of human needs. The market system, although self-interested, was more impersonal. It sent foresters “long distance” expressions of forest social values. Public hearings, client interaction, or political demonstrations bring forest social values face-to-face with foresters. It has generally made forest management more personal and, to many foresters, less comfortable.

In failing to view their role as managing for changing forest social values, foresters were often shocked and reactionary to socio-political communication from their citizens that forest priorities and management practices should change. Sometimes the results of such resistance to social changes were large-scale conflicts over forest management practices in many western nations, which is the subject of the next section.

FACTORS UNDERLYING NATIONAL CONFLICTS OVER FOREST SOCIAL VALUES

In much multiple-use forest management, wood production coexists with picnicking, landscape or wildlife uses. Some adjustment in the management of each allows multiple forest values to be accommodated with minimal conflict. But in many western countries forest issues have escalated

to national conflicts, making front-page newspaper headlines — the harvesting of Swedish beech (*Fagus sylvatica*) or New Zealand native forests, for example. Examining the underlying factors common to these heated forestry issues might provide lessons to help minimize future conflicts.

Common ingredients in most national forestry conflicts appear to be:

(1) an urban society that appreciates romantic, non-utilitarian forest or wildlife values, and has power to implement its views through effective socio-political action;

(2) a perception that forest and wildlife have a prominent, sentimental and nostalgic role in a nation's heritage; and

(3) identifiable forestry professional groups and organizations that tend to be reluctant to trade-offs of wood production for other forest uses. Each one of these is examined below.

An urban society with romantic, non-product forest values

Many agricultural children are introduced to forest values with an axe or gun in their hands (Hendee, 1969; Kennedy, 1973). They learn to value plants and animals primarily for their ability to satisfy human needs, especially monetary needs. Trees are good for timber or fuel. Some animals are good (game) and other bad (vermin or predators). This practical, utilitarian perception of nature is reinforced in logging, farming or mining employment, and in common rural recreational pursuits like hunting, fishing or trapping (see Table 1). Studies like those of Buttell and Flinn (1974) or Lowe and Pinhey (1982) require a caveat in using terms like rural society to generalize about nature values. Their rural samples in agricultural occupations had less concern for environmental problems than non-farm/rural residents (even when controlled for education). They found non-farm/rural residents were as concerned about environmental problems as urbanites. The studies examined broad environmental and pollution concerns and did not focus on specific forest resource issues. Thus the association of nonconsumptive forest values with agricultural/rural society and non-consumptive values with urban society is employed here as a reasonable hypothesis with considerable empirical support, but more validation is needed.

It was in a practical, agricultural/rural value system that the American conservation movement, the forester profession and forest resource agencies were largely created (Hays, 1959). Although the forest conservation movement centered on utilitarian values of current and future generations, it also drew considerable support from eastern American urbanites — many with strong non-utilitarian concerns (Gilligan, 1954; Nash, 1967). Social conflict over the damming of the Hetch-Hetchy Canyon became a national issue between 1908–13, and the strength of American nonconsumptive/symbolic nature values was later expressed in political action in the establishment of the National Park Service (1916). The USDA Forest Service

TABLE 1

A depiction of two often-conflicting sets of forest/nature uses and values

Issues	Agricultural/rural orientation to nature	Urban, orientation to nature
1. Learning the value of forests, wild animals or nature	Cutting firewood Farming Weeding fields Protecting crops from disease, insects, crows	Watching television or Walt Disney nature films Reading books Forest recreation experience (i.e., camping, etc.)
2. Popular outdoor recreational activities	Hunting Fishing Trapping	Hiking and nature study Camping, auto-touring Bird-watching rather than hunting
3. Dominant values/attitudes toward forests or nature	Practical, utilitarian values dominate Economic worth of things Focus is on material and instrumental values	Romantic, idealistic values Nature does not have to be used or sold to have value These values are often viewed as impractical, unrealistic or "feminine" by rural society
4. Dependence on nature to earn a livelihood	Agriculture, logging or mining directly utilizing nature Income, fuelwood, food and recreation comes directly from forest and nature	Professional, clerical, manufacturing, and other types of jobs that often do not use natural resources in their raw (or natural) condition Their jobs are distant from and indirectly dependent on using nature

responded to such nonconsumptive values and political competition with expanded recreational programs, plus the creation of wilderness and primitive areas (Baldwin, 1972; Tweed, 1980). By the 1960s such non-utilitarian values were widespread, and Americans (largely urban) increasingly challenged the view that economic and utilitarian benefits were the most legitimate indicator of forest and wildlife worth (Table 1). Urbanites had the luxury of being able to focus on romantic, idealistic forest values, for, unlike their agricultural ancestors, few directly utilized nature for a livelihood.

Social conflicts in North America over harvesting redwood forests or classifying forest areas as wilderness are largely clashes between utilitarian and romantic forest values. The conversion of Australian and New Zealand native forest or Scottish moorland to exotic conifer plantations generated

similar conflicts. These social conflicts illustrate how professional foresters often fail to appreciate the *symbolic* value of natural resources (Reunala, 1984); and they are evidence of how inadequately commodity prices and potential rural employment express the full social value of some forest resources today. To many Americans, it is as realistic to estimate the social value of the Liberty Bell in its brass salvage price, or the social value of New York's Central Park in potential grazing fees, as it is to estimate old-growth redwood values in their market prices or number of potential homes that could be constructed.

Nostalgic, heritage values of forests and wildlife

Countries like the United States, Germany and Sweden have a nostalgic, nationalistic imagery of pristine forests and wild animals. The urban New Zealand decedents of Maori and Europeans also have cherished images of the nation's tall forests and native birds of 100 years ago. New Zealand foresters learned that some native forests they planned to harvest were perceived as museum forests or national heirlooms — not just a source of wood, rural employment, or foreign exchange values (Kennedy, 1981).

California redwood forests, Swedish beech forests, or Australian *Eucalyptus* spp. forests symbolize to many of their citizens a pristine past — a past when natural forces dominated, life was simpler, and humans were less driven by crass economic motives (so the folklore goes). Like the harp seal pup, these physical entities often take on a potent symbolic value that can galvanise strong and widespread resistance when forest managers focus primarily on the economic values of these forests.

Countries like the Republic of Ireland have little nostalgic, nationalistic forest heritage. Such a missing ingredient may explain why Ireland has experienced less social conflict over forest management than other European countries (Kennedy and McCusker, 1983). This may change as generations of Irish are raised around exotic State Forests and increasingly consider them a normal part of their life, landscape, and national heritage.

Defensive, reactionary forestry professionals and/or agencies

It takes at least two opposing forces to generate social conflicts worthy of newspaper headlines.

Foresters tend to be a proud, cohesive professional group. Their initial responses to individuals and groups promoting recreational, wildlife, and landscape values in the last few decades were often defensive and reactionary. In the United States, the Forest Service and Society of American Foresters did not support (or resisted) passage of the Wilderness Act (1964) and the National Environmental Policy Act (1970), both overwhelmingly approved by Congress. Such resistance encouraged American environmental groups, like the Sierra Club, to adopt a combative stance toward foresters.

These environmental groups grew in membership and became more militant in character.

The foresters' response to "non-professional" environmental groups was not surprising. Like engineers and physicians, foresters tend to discount advice and criticism from those not trained in their discipline — and many forestry critics had never labored through a silviculture course. Given such attitudes, it is not surprising that the courts have become a common setting in which American forestry values are debated and decided. American foresters, like other professionals, were also caught up in an unexpected reversal in social attitudes that accorded less respect to scientifically trained professionals than in earlier decades.

A decade after the National Environmental Policy Act (1970) and other "environmental" legislation, American National Forests now begin their planning process by identifying important *social issues*. They then seek to resolve these conflicts in a Forest Plan that many citizens consider a "social contract" (Shannon, 1984) between the public and agency professionals — setting approved goals and management options to be pursued over 10-year periods.

CLOSING COMMENT

If one accepts the premise that foresters manage resources to accommodate immediate and long-run social values, and that many forest social values conflict with one another, then foresters can be viewed as conflict managers. In what they do (and fail to do) foresters can usually intensify or dampen social conflict over forest values.

Most forestry students, I would hypothesize, do not select a career in forestry to be conflict managers. Quite the contrary. Many surveys of forestry students, from Eyre's (1929) study of 1928 forestry graduates to Kennedy and Mincolla's study (1982), indicate that foresters are attracted to the perceived simplicity and tranquility of working in and with forests. The stress and ambiguity of dealing with social issues often intimidate foresters. American college students who plan to deal with social conflict usually become lawyers, social workers, or school psychologists. Forestry students generally seek the simplicity and solace of forests and rural communities (Holland and Beazley, 1971). The forest recreation boom of the early 1960s dragged many American foresters, resisting doggedly, into twentieth century forest management (Gould, 1962; Wagar, 1975). In previous decades, social values were largely accommodated by protecting forests or supervising the process of getting logs on their way to distant consumers. Dealing with forest recreationists, conducting public hearings, or participating in special interest group meetings has made forest management a more personal, face-to-face endeavor. Many foresters, like myself, probably selected their profession to avoid heavy human interaction and confrontation in the normal workday. Long-distance, impersonal, "objec-

tive" lumber prices in quarterly reports were a more preferred and less stressing means of communicating social values.

After a B.S. degree based on science and courses that focused on silviculture and forest protection, I initially thought foresters managed physical things. It requires deeper and more detached thinking to look through these things and see the social values behind them. To some in our profession, traditional forest management is so narrowly associated with wood products and market prices that the concept of "social forestry" had to be invented to emphasize professional responsibility to broad and varied social needs — like those of the Ikalahan forest dwellers of the Philippines (Rice, 1984). Its indictment of our narrow value focus that a new term such as social forestry had to be created to draw attention to what forestry has always been: an endeavor to provide professionally sound information and forest management alternatives to accommodate a variety of current social values, while maintaining options for future generations. That has always been the role of foresters and, as long as society will have us, it always will.

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